AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS X

MODEL EXAMINATION PAPER 2027 AND ONWARDS

Chemistry Paper II

Time: 1 hour 50 minutes Marks: 25

This model paper aims to represent the table of specification with regards to the topics/ content and cognitive demand only.

AKU-EB Paper 2027 for Teaching Learning only

INSTRUCTIONS:

1. Check your name and school information above. Sign if the information is correct.

I agree that this is my name and school. **Candidate's Signature**

2. Read each question carefully.
3. There are SIX questions in this exam. Answer ALL questions. Questions 5 and 6 each offer TWO choices. For each of these questions, attempt ONE choice only.
4. When answering the questions:
   - Use a black pen/ pointer to write your answers. DO NOT write your answers in pencil.
   - Use a black pencil for diagrams and drawings. DO NOT use coloured pencils, markers or highlighters.
   - DO NOT use staples, paper clips, glue, correcting fluid, or ink erasers.
     If you make a mistake, cross it out neatly and write the correct answer beside it.
   - Write your answers only in the allocated space provided for each question.
   - DO NOT write outside the answer space. Answers written outside the allocated space will not be marked.
5. The marks for each question are shown in brackets ( ) next to the question number. Use this information to manage your time efficiently.
6. DO NOT mention your name or school name anywhere inside the paper.
7. You may use a simple calculator if you wish.

Q.1. (Total 4 Marks)

a. Consider the following chemical equation.

$2H_{2(g)} + O_{2(g)} \rightleftharpoons 2H_2O_{(l)}$

i. Identify the direction in which the given equation shows endothermic reaction. (1 Mark)

_____

_____

ii. Give ONE reason to justify your answer to part i. (1 Mark)

_____

_____

b. Consider the given reaction at equilibrium.

$CO_{(g)} + H_2O_{(g)} \rightleftharpoons CO_{2(g)} + H_{2(g)}$

What will be the effect on the concentration of H_2O if a small amount of CO gas is added to the reaction mixture? Give a reason to support your answer. (2 Marks)

_____

_____

_____

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Q.2. (Total 3 Marks)

a. What is meant by the term ‘homologous series’? (1 Mark)

_____

_____

b. State the general formula for alkanes and alkynes. (2 Marks)

Alkanes: _____

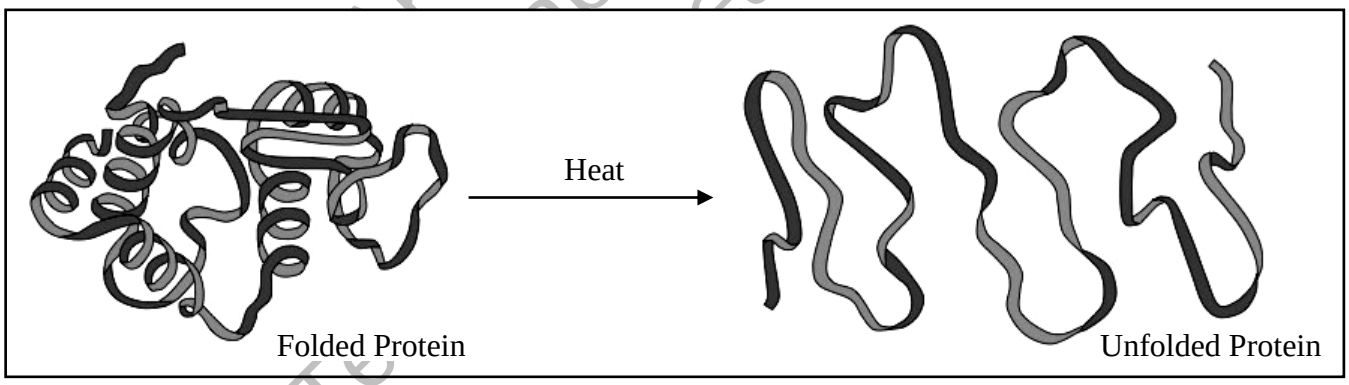
_____

Alkynes: _____

_____

Q.3. (Total 3 Marks)

The given diagram illustrates the change in the structure of a protein molecule due to increase in temperature.



a. Name the process of changing a folded protein molecule into an unfolded protein molecule. (1 Mark)

_____

b. Write ONE chemical change that causes the unfolding of the protein molecule. (1 Mark)

_____

_____

c. What is the effect of unfolding on the function of the protein molecule? (1 Mark)

_____

_____

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Q.4. (Total 3 Marks)

An industrial plant receives hard water containing calcium and magnesium ions. The plant wants to make the water soft using an ion exchange resin.

a. Identify the type of hardness in water that will be removed by the ion exchanger. (1 Mark)

_____

_____

b. Describe the process of removing this hardness from water through the ion exchange resin. (1 Mark)

_____

_____

c. If the resin becomes saturated, suggest a practical method to regenerate it so that it can be reused. (1 Mark)

_____

_____

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Q.5. (Total 6 Marks)

EITHER

a. In a titration process, 15 cm^3 of 0.5 M dilute sulphuric acid is completely used to neutralise 10 cm^3 of sodium hydroxide solution.

Calculate the molarity of sodium hydroxide solution with the help of a balanced chemical equation. (6 Marks)

OR

b.

i. Using balanced chemical equation(s), illustrate the complete steps of oxidation of the following organic compounds with potassium permanganate ($KMnO_4$) solution. (4 Marks)

I. Ethene

II. Ethyne

ii. Name the final products obtained on complete oxidation of (I) ethene and (II) ethyne in part i. (2 Marks)

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Q.6. (Total 6 Marks)

EITHER

a. When petrol is burnt in an internal combustion engine of a car by an electrical spark which passes through the air/ petrol mixture, it produces gases such as carbon monoxide and oxides of nitrogen. The release of these gases from car exhaust considerably pollutes the environment.

Evaluate which gas poses a greater threat to the environment in urban areas of Pakistan by

i. identifying the main pollution problem caused by each of these gases. (2 Marks)

ii. analysing any TWO harmful impacts of each of the identified problems (in part i) on the environment. (4 Marks)

OR

b. A blast furnace is charged with haematite ore, coke and limestone from the top. It is then heated by blowing hot air in it from the base.

Using the given information, evaluate the effectiveness of the stages that enable the extraction of iron from its ore. Write chemical equations and/ or reasons to support your answer. (6 Marks)

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END OF PAPER

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